OVERVIEW PROSES

INPUT: Data OHLCV saham
↓
STEP 1: Deteksi Pivot Points (Local Extrema)
↓
STEP 2: Filter berdasarkan Current Price
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STEP 3: Standardisasi Data (Z-score Normalization)
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STEP 4: K-Means Clustering
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STEP 5: Penentuan Range Zona
↓
STEP 6: Scoring & Selection
↓
OUTPUT: 1 Support Zone + 1 Resistance Zone

STEP 1: DETEKSI PIVOT POINTS

Tujuan:

Mengidentifikasi titik-titik pembalikan harga (reversal points) yang merupakan kandidat Support dan Resistance.

Metode: Local Extrema Detection

Algoritma:

```
pivot_high = argrelextrema(High, np.greater_equal, order=5)
pivot_low = argrelextrema(Low, np.less_equal, order=5)
```

Definisi Matematis:

Pivot High di index i jika:

$H(i) \geq H(i-n), H(i-n+1), ..., H(i-1), H(i+1), ..., H(i+n)$

Pivot Low di index i jika:

$L(i) \leq L(i-n), L(i-n+1), ..., L(i-1), L(i+1), ..., L(i+n)$

Dimana:

- $H(i)$ = harga High pada candle ke-i
- $L(i)$ = harga Low pada candle ke-i
- n = order (default = 5)

Contoh Perhitungan:

Data harga High 7 candle berturut-turut:

Index: [0] [1] [2] [3] [4] [5] [6]
High: 9500 → 9700 → 9900 → 9800 → 9600 → 9400 → 9300

Dengan order=2, cek apakah index [2] adalah pivot high:

- H[2] = 9900
- Kiri: H[0]=9500, H[1]=9700
- Kanan: H[3]=9800, H[4]=9600

Kondisi:

- 9900 ≥ 9500
- 9900 ≥ 9700
- 9900 ≥ 9800
- 9900 ≥ 9600

Kesimpulan: Index [2] adalah **Pivot High** (potential resistance)

Referensi:

1. **De Prado, M. L. (2018).** *Advances in Financial Machine Learning*. Wiley.
    - Chapter 17, Section 17.3: "Structural Breaks" - membahas peak/trough detection menggunakan local extrema.
2. **Murphy, J. J. (1999).** *Technical Analysis of the Financial Markets*. New York Institute of Finance.
    - Chapter 4: "Support and Resistance" - menjelaskan bahwa S/R terbentuk di titik-titik swing high/low.

Output STEP 1:

Misalkan dari 100 candle terdeteksi:

- **Pivot Highs:** [10500, 10600, 10550, 10700, 10650, 11000, 10900]
- **Pivot Lows:** [9000, 9100, 9050, 8900, 8500, 8600, 9200]

STEP 2: FILTER BERDASARKAN CURRENT PRICE

Tujuan:

Memastikan Support di bawah harga saat ini, dan Resistance di atas harga saat ini.

Logika:

Misalkan **Current Price = 9850**

Support Filter:

S = {p ∈ Pivot Lows | p < P_current}

Resistance Filter:

R = {p ∈ Pivot Highs | p > P_current}

Contoh Perhitungan:

Before Filter:

- Pivot Lows: [9000, 9100, 9050, 8900, 8500, 8600, 9200]
- Pivot Highs: [10500, 10600, 10550, 10700, 10650, 11000, 10900]
- Current Price: 9850

After Filter:

Support Levels (< 9850):

9000 , 9100 , 9050 , 8900 , 8500 , 8600 , 9200

Semua lolos karena < 9850

Resistance Levels (> 9850):

10500 , 10600 , 10550 , 10700 , 10650 , 11000 , 10900

Semua lolos karena > 9850

Justifikasi:

Ini adalah **prinsip dasar Support & Resistance** dari technical analysis:

- Support = level di mana harga cenderung "terpental ke atas" → harus di BAWAH harga saat ini
- Resistance = level di mana harga cenderung "terpental ke bawah" → harus di ATAS harga saat ini

Referensi:

- **Osler, C. L. (2003).** "Currency orders and exchange rate dynamics: An explanation for the predictive success of technical analysis." *The Journal of Finance*, 58(5), 1791-1819.
    - Menjelaskan bahwa S/R terbentuk karena clustering of limit orders di level-level tertentu.

STEP 3: STANDARDISASI DATA (Z-SCORE NORMALIZATION)

Tujuan:

Menormalisasi data agar K-Means algorithm dapat bekerja optimal.

Formula:

$z = (x - \mu) / \sigma$

Dimana:

- x = nilai asli (harga)
- μ = mean dari data
- σ = standard deviation
- z = nilai ter-standardisasi

Contoh Perhitungan:

Support Levels: [9000, 9100, 9050, 8900, 8500, 8600, 9200]

Step 1: Hitung Mean (μ)

$\mu = (9000 + 9100 + 9050 + 8900 + 8500 + 8600 + 9200) / 7$
$\mu = 62350 / 7$
$\mu = 8907.14$

Step 2: Hitung Standard Deviation (σ)

Variance:

$\sigma^2 = \Sigma(x\_i - \mu)^2 / n$

Perhitungan:

$(9000 - 8907.14)^2 = 8618.74$
$(9100 - 8907.14)^2 = 37192.34$
$(9050 - 8907.14)^2 = 20408.34$
$(8900 - 8907.14)^2 = 50.98$
$(8500 - 8907.14)^2 = 165743.14$
$(8600 - 8907.14)^2 = 94294.94$
$(9200 - 8907.14)^2 = 85785.74$

$\sigma^2 = 412094.22 / 7 = 58870.60$
$\sigma = \sqrt{58870.60} = 242.63$

Step 3: Standardize Setiap Nilai

$z_1 = (9000 - 8907.14) / 242.63 = 0.383$
$z_2 = (9100 - 8907.14) / 242.63 = 0.795$
$z_3 = (9050 - 8907.14) / 242.63 = 0.589$
$z_4 = (8900 - 8907.14) / 242.63 = -0.029$
$z_5 = (8500 - 8907.14) / 242.63 = -1.678$
$z_6 = (8600 - 8907.14) / 242.63 = -1.266$
$z_7 = (9200 - 8907.14) / 242.63 = 1.207$

Hasil Standardisasi:

Original: [9000, 9100, 9050, 8900, 8500, 8600, 9200]
Z-score: [0.38, 0.80, 0.59, -0.03, -1.68, -1.27, 1.21]

Referensi:

1. **Hastie, T., Tibshirani, R., & Friedman, J. (2009).** *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd ed.). Springer.
    - Section 14.5.1: "Standardization and Distance Measures" - menjelaskan pentingnya standardisasi untuk clustering algorithms.
2. **Pedregosa, F., et al. (2011).** "Scikit-learn: Machine Learning in Python." *Journal of Machine Learning Research*, 12, 2825-2830.
    - Dokumentasi StandardScaler implementation.

Kenapa Perlu Standardisasi?

1. K-Means menggunakan Euclidean Distance yang sensitive terhadap scale
2. Tanpa standardisasi, fitur dengan nilai besar akan mendominasi
3. Convergence algorithm lebih cepat dan stabil

STEP 4: K-MEANS CLUSTERING

Tujuan:

Mengelompokkan pivot points yang berdekatan menjadi zona Support/Resistance.

Algoritma K-Means:

Objective Function:

min Σ(i=1 to k) Σ(x∈C_i) ||x - μ_i||²

Dimana:

- k = jumlah cluster (default = 3)
- C_i = cluster ke-i
- μ_i = centroid cluster ke-i
- ||x - μ_i||² = squared Euclidean distance

Algoritma:

1. Initialize k centroids secara random (atau k-means++)
2. Repeat until convergence:
   a. Assign setiap point ke centroid terdekat

b. Update centroid = mean dari semua points di cluster
3. Return final clusters

Contoh Perhitungan Detail:

Data (Standardized): [0.38, 0.80, 0.59, -0.03, -1.68, -1.27, 1.21]

Kita ingin **k = 2 cluster**

Iteration 0: Initialize Centroids

- C1 = 0.38 (random pick pertama)
- C2 = -1.68 (random pick kedua)

Iteration 1:

Step A: Assign Points

Point 1 (0.38):

- Distance to C1: |0.38 - 0.38| = 0.00
- Distance to C2: |0.38 - (-1.68)| = 2.06
- **Assigned to: C1** (closer)

Point 2 (0.80):

- Distance to C1: |0.80 - 0.38| = 0.42
- Distance to C2: |0.80 - (-1.68)| = 2.48
- **Assigned to: C1**

Point 3 (0.59):

- Distance to C1: |0.59 - 0.38| = 0.21
- Distance to C2: |0.59 - (-1.68)| = 2.27
- **Assigned to: C1**

Point 4 (-0.03):

- Distance to C1: |-0.03 - 0.38| = 0.41
- Distance to C2: |-0.03 - (-1.68)| = 1.65
- **Assigned to: C1**

Point 5 (-1.68):

- Distance to C1: |-1.68 - 0.38| = 2.06
- Distance to C2: |-1.68 - (-1.68)| = 0.00

- **Assigned to: C2**

Point 6 (-1.27):

- Distance to C1: |-1.27 - 0.38| = 1.65
- Distance to C2: |-1.27 - (-1.68)| = 0.41
- **Assigned to: C2**

Point 7 (1.21):

- Distance to C1: |1.21 - 0.38| = 0.83
- Distance to C2: |1.21 - (-1.68)| = 2.89
- **Assigned to: C1**

Result Iteration 1:

Cluster 1: [0.38, 0.80, 0.59, -0.03, 1.21] → 5 points
Cluster 2: [-1.68, -1.27] → 2 points

Step B: Update Centroids

New C1:

μ_1 = (0.38 + 0.80 + 0.59 + (-0.03) + 1.21) / 5
μ_1 = 2.95 / 5
μ_1 = 0.59

New C2:

μ_2 = (-1.68 + (-1.27)) / 2
μ_2 = -2.95 / 2
μ_2 = -1.475

Iteration 2:

(Repeat assignment dan update sampai convergence...)

Final Clusters (setelah convergence):

Cluster 1 (Upper Support): [9000, 9100, 9050, 8900, 9200]

- Centroid (standardized): 0.59
- Centroid (original): 9050

Cluster 2 (Lower Support): [8500, 8600]

- Centroid (standardized): -1.475
- Centroid (original): 8550

Transform Back ke Harga Asli:

$x = z \times \sigma + \mu$

Contoh untuk Cluster 1 centroid:

$x = 0.59 \times 242.63 + 8907.14$
$x = 143.15 + 8907.14$
$x = 9050.29$

Referensi:

1. **MacQueen, J. (1967).** "Some methods for classification and analysis of multivariate observations." *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, 1(14), 281-297.
   - **Paper ORIGINAL K-Means Algorithm**
2. **Arthur, D., & Vassilvitskii, S. (2007).** "k-means++: The advantages of careful seeding." *Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms*, 1027-1035.
   - Improved initialization method (digunakan oleh sklearn)

STEP 5: PENENTUAN RANGE ZONA

Tujuan:

Mengubah cluster points menjadi zona dengan range (min-max).

Formula:

Untuk setiap cluster C_i:

$Zone_i = [\min(C_i) - m, \max(C_i) + m]$

Dimana margin:

$m = (\max(C_i) - \min(C_i)) \times 0.1$

Contoh Perhitungan:

Cluster 1: [9000, 9100, 9050, 8900, 9200]

Step 1: Find Min & Max

Min = 8900
Max = 9200

Step 2: Calculate Margin

$m = (9200 - 8900) \times 0.1$
$m = 300 \times 0.1$
$m = 30$

Step 3: Final Zone Range

$Zone_1 = [8900 - 30, 9200 + 30]$
$Zone_1 = [8870, 9230]$

Cluster 2: [8500, 8600]
Min = 8500
Max = 8600
Margin = $(8600 - 8500) \times 0.1 = 10$

$Zone_2 = [8500 - 10, 8600 + 10]$
$Zone_2 = [8490, 8610]$

Kenapa Pakai Range (bukan garis exact)?

Support dan Resistance dalam praktik **bukan level harga yang presisi**, melainkan **area price zones**.

Referensi:

1. **Murphy, J. J. (1999).** *Technical Analysis of the Financial Markets*. New York Institute of Finance.

    ○ Chapter 4, halaman 53: "Support and resistance should be thought of as **zones** rather than precise levels."
2. **Osler, C. L. (2003).** "Currency orders and exchange rate dynamics." *The Journal of Finance*, 58(5), 1791-1819.

    ○ Empirical evidence menunjukkan S/R terbentuk sebagai price zones karena clustering of orders.

Strength Calculation:

Strength_i = |C_i|

(Jumlah pivot points dalam cluster)

Contoh:

- Cluster 1: Strength = 5 (5 pivot points)
- Cluster 2: Strength = 2 (2 pivot points)

Referensi:

- **Brock, W., Lakonishok, J., & LeBaron, B. (1992).** "Simple technical trading rules and the stochastic properties of stock returns." *The Journal of Finance*, 47(5), 1731-1764.
    - ○ Menunjukkan bahwa S/R yang di-test berkali-kali (multiple touches) lebih reliable.

STEP 6: SCORING & SELECTION

Tujuan:

Memilih 1 zona Support dan 1 zona Resistance yang PALING VALID berdasarkan proximity dan strength.

Formula Scoring:

Score = n / (d + ε)

Dimana:

- n = Strength (jumlah touch points)

- d = Normalized distance
- ε = 0.01 (untuk avoid division by zero)

Normalized Distance:

Untuk Support:

d_support = |P_current - Zone_max| / P_current

Untuk Resistance:

d_resistance = |Zone_min - P_current| / P_current

Contoh Perhitungan Lengkap:

Misalkan **Current Price = 9850**

Support Zones yang terdeteksi:

Zone A:

- Range: [8870, 9230]
- Strength: 5

Distance:

d_A = |9850 - 9230| / 9850
d_A = 620 / 9850
d_A = 0.0629

Score:

Score_A = 5 / (0.0629 + 0.01)
Score_A = 5 / 0.0729
Score_A = 68.58

Zone B:

- Range: [8490, 8610]
- Strength: 2

Distance:

d_B = |9850 - 8610| / 9850
d_B = 1240 / 9850
d_B = 0.1259

Score:

Score_B = 2 / (0.1259 + 0.01)
Score_B = 2 / 0.1359
Score_B = 14.72

Comparison:

Zone	Range	Strength	Distance	Score
A	[8870, 9230]	5	6.29%	**68.58**
B	[8490, 8610]	2	12.59%	14.72

Kesimpulan: Zone A dipilih sebagai **Support yang paling valid** karena:

1. Lebih dekat dengan harga saat ini (6.29% vs 12.59%)
2. Lebih banyak touch points (5 vs 2)
3. Score lebih tinggi (68.58 vs 14.72)

Referensi Multi-Criteria Scoring:

1. **Saaty, T. L. (2008).** "Decision making with the analytic hierarchy process." *International Journal of Services Sciences*, 1(1), 83-98.
    - Multi-criteria decision making framework untuk kombinasi multiple factors.
2. **Triantaphyllou, E. (2000).** *Multi-criteria Decision Making Methods: A Comparative Study*. Springer.
    - Weighted scoring methods untuk optimization problems.

Final Output:

SUPPORT: Rp 8,870 - Rp 9,230
  Touch points: 5
  Score: 68.58

RESISTANCE: Rp 10,480 - Rp 10,720
  Touch points: 6

Score: 75.32

RINGKASAN MATEMATIS LENGKAP

1. Pivot Detection:

$P\_high(i) = \{i \mid H(i) \geq H(i-n), ..., H(i+n)\}$
$P\_low(i) = \{i \mid L(i) \leq L(i-n), ..., L(i+n)\}$

2. Filtering:

$S = \{p \in P\_low \mid p < P\_current\}$
$R = \{p \in P\_high \mid p > P\_current\}$

3. Standardization:

$z = (x - \mu) / \sigma$

4. K-Means:

$\min \Sigma(i=1 \text{ to } k)\ \Sigma(x \in C\_i)\ ||x - \mu\_i||^2$

5. Zone Range:

$Zone = [\min(C) - 0.1(\max(C) - \min(C)), \max(C) + 0.1(\max(C) - \min(C))]$

6. Scoring:

Score = Strength / (Normalized Distance + 0.01)

DAFTAR REFERENSI LENGKAP

Primary References (Core Algorithm):

1. **MacQueen, J. (1967).** "Some methods for classification and analysis of multivariate observations." *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics*

and Probability, 1(14), 281-297.

2. **Hastie, T., Tibshirani, R., & Friedman, J. (2009).** *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd ed.). Springer.

3. **Murphy, J. J. (1999).** *Technical Analysis of the Financial Markets: A Comprehensive Guide to Trading Methods and Applications*. New York Institute of Finance.

Supporting References:

4. **De Prado, M. L. (2018).** *Advances in Financial Machine Learning*. Wiley.

5. **Arthur, D., & Vassilvitskii, S. (2007).** "k-means++: The advantages of careful seeding." *Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms*, 1027-1035.

6. **Osler, C. L. (2003).** "Currency orders and exchange rate dynamics: An explanation for the predictive success of technical analysis." *The Journal of Finance*, 58(5), 1791-1819.

7. **Brock, W., Lakonishok, J., & LeBaron, B. (1992).** "Simple technical trading rules and the stochastic properties of stock returns." *The Journal of Finance*, 47(5), 1731-1764.

8. **Saaty, T. L. (2008).** "Decision making with the analytic hierarchy process." *International Journal of Services Sciences*, 1(1), 83-98.

9. **Pedregosa, F., et al. (2011).** "Scikit-learn: Machine Learning in Python." *Journal of Machine Learning Research*, 12, 2825-2830.
